

Sri Lanka Institute of Information Technology



Streetify

Web-Based Transportation System.

Lab Report 02: Agile Development (Scrum Framework)

Group ID : 2026-Y2-S1-KU-24 (Group 24)

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Introduction

Project Overview

Streetify is an innovative, web-based transportation management platform designed specifically for the Sri Lankan urban transit ecosystem (developed by Group 24 of Batch 1 SLIIT Kandy Uni). Inspired by global ride-hailing services like Uber, Streetify provides a localized, end-to-end digital solution that seamlessly connects passengers, drivers, and administrative operational teams within a single web application.

Users of the System (Personas)

- ❖ Daily Passenger & Driver
- ❖ Senior Driver Coordinator
- ❖ System Administrator
- ❖ Operations Manager
- ❖ Finance & Payments Manager
- ❖ Customer Experience Officer

Client Scenario: SwiftRide Lanka (Pvt) Ltd

Functional Requirements (User Needs)

User Authentication & Roles

- ❖ Users should register/login using credentials.
- ❖ System should assign roles (Passenger, Driver, System Admin, Finance Staff, Customer Support) with appropriate permissions.

Ride Search & Fare Estimation

- ❖ Passengers can enter pickup and destination locations.
- ❖ System calculates and displays estimated fares based on route and distance before booking.

Ride Booking & Tracking

- ❖ Passengers can request a ride and match with available nearby drivers.
- ❖ System displays driver details, vehicle information, and ride status.

Trip Execution & Management

- ❖ Drivers can accept or reject incoming ride requests.
- ❖ Drivers update trip status (Arrived, Trip Started, Completed, Canceled).

Payment & Receipt Processing

- ❖ Passengers can pay using card, digital wallet, or cash.
- ❖ Automated digital receipts are generated upon trip completion.

Notifications & Reminders

- ❖ Automated notifications for ride request updates, payment confirmation, and cancellation alerts.

Passenger & Driver Dashboard

- ❖ View ride history, transaction details, active ride status, and ratings.
- ❖ Drivers view daily trip records and total earnings.

Review, Rating & Support Dashboard

- ❖ Passengers and drivers can submit star ratings and written reviews after rides.
- ❖ Customer support officers investigate passenger complaints and cancellation disputes.

Administrative & Finance Control

- ❖ Admins verify pending driver registration documents and monitor real-time system activities.
- ❖ Finance managers track commissions, manage driver earnings, and process refunds.

System Reporting & Settings

- ❖ Admins and finance staff can generate operational and financial reports (trip volume, user activity, total revenue).
- ❖ Configure system settings, policy enforcement rules, account deactivation, and commission rates.

Scrum Roles

Scrum Role	Student ID	Member Assigned	Co-ownership & Responsibilities
Product Owner (Frontend/Backend Developer)	IT25102240	Rammandalagedara R.V.S. (Admin & System Control Lead)	Administration & System Control & Backlog Acceptance
Scrum Master (Frontend/Backend Developer)	IT25102208	Nayanamina A.R.L.P. (User Account Lead)	User Account Management & Scrum Facilitation
Developer 1 (Frontend/Backend Developer)	IT25102207	Dharmakeerthi W.A.C.B.	Ride Booking & Management.
Developer 2 (Frontend/Backend Developer)	IT25102241	Senaka K.A.T.	Driver Trip Request Management.
Developer 3 (Database & Integration Developer)	IT25102225	Edirisinghe E.A.R.N.D.	Payment & Receipt Management.
Developer 4 (UI/UX & QA Developer)	IT25102193	Weerasingha W.A.M.B.	Review, Rating & Feedback Management.

Product Owner: owns value ordering, clarifies acceptance criteria, and accepts finished work

Scrum Master: facilitates Scrum events, makes blockers visible, and helps the team follow the agreed process.

Developers: build, test, integrate, and demonstrate selected sprint work as a cross-functional team.

Ownership rule: *Scrum accountability does not replace the CRUD division at the requirement gathering phase. Nayanamina A.R.L.P. retains User Account Management, Dharmakeerthi W.A.C.B. retains Ride Booking and Management, Senaka K.A.T. retains Driver Trip Request Management, Edirisinghe E.A.R.N.D. retains Payment and Receipt Management, Weerasingha W.A.M.B. retains Review, Rating, and Feedback Management, and Rammandalagedara R.V.S. retains Administration and System Control. A functional owner defines domain rules and verifies behaviour, Developers implement and test, only the Product Owner accepts a completed backlog item.*

Identified Personas with User Stories

As required by Lab 01 & 02 instructions, here are the **6 Personas** (derived from client representatives) with **4 User Stories each** (24 user stories total)

- ❖ Daily Passenger and Driver (System User)
- ❖ Senior Driver Coordinator (Mr. Kasun Jayawardena)
- ❖ System Administrator (Mr. Ravindu Senanayake)
- ❖ Operations Manager (Ms. Nadeesha Perera)
- ❖ Finance & Payments Manager (Ms. Dilini Wickramasinghe)
- ❖ Customer Experience Officer (Ms. Tharushi Fernando)

1. Daily Passenger and Driver (System User)

User Story ID	User Story
US 1.1	As a Daily Passenger, I want to create an account and log in securely so that I can access ride-booking services.
US 1.2	As a Daily Passenger, I want to enter my pickup and drop-off points to view estimated fares before booking.
US 1.3	As a Driver, I want to receive real-time ride notifications so that I can accept or decline incoming trip requests.
US 1.4	As a Driver, I want to update trip progress (<i>Arrived, Started, Completed</i>) so that the system logs my ride activity.

2. Senior Driver Coordinator (Mr. Kasun Jayawardena)

User Story ID	User Story
US 2.1	As a Senior Driver Coordinator, I want to inspect driver registration applications so that I can ensure only qualified drivers join the fleet.
US 2.2	As a Senior Driver Coordinator, I want to verify driver license and vehicle documents so that platform compliance is strictly maintained.
US 2.3	As a Senior Driver Coordinator, I want to review driver performance metrics and ratings so that high-performing drivers are recognized.
US 2.4	As a Senior Driver Coordinator, I want to monitor repeated trip cancellations by drivers so that appropriate operational warnings can be issued.

3. System Administrator (Mr. Ravindu Senanayake)

User Story ID	User Story
US 3.1	As a System Admin, I want to configure role-based permissions so that users access features relevant to their roles.
US 3.2	As a System Admin, I want to monitor real-time system activity dashboards so that I can ensure high application availability.
US 3.3	As a System Admin, I want to suspend accounts that commit severe policy violations so that security standards remain high.
US 3.4	As a System Admin, I want to view system audit logs so that platform activity can be analyzed during technical reviews.

4. Operations Manager (Ms. Nadeesha Perera)

User Story ID	User Story
US 4.1	As an Operations Manager, I want to view daily trip demand statistics so that regional ride allocation can be planned effectively.
US 4.2	As an Operations Manager, I want to track active driver availability trends during peak commuting hours so that service gaps are minimized.
US 4.3	As an Operations Manager, I want to inspect overall trip cancellation rates so that operational bottlenecks can be resolved.
US 4.4	As an Operations Manager, I want to generate operational reports so that performance insights can be shared with management.

5. Finance & Payments Manager (Ms. Dilini Wickramasinghe)

User Story ID	User Story
US 5.1	As a Finance Manager, I want to review transaction logs across card, digital wallet, and cash payments so that financial records are complete.
US 5.2	As a Finance Manager, I want to track company commissions deducted from completed trips so that revenue calculation is accurate.
US 5.3	As a Finance Manager, I want to process approved refund requests for failed or disputed transactions so that client accounts stay balanced.
US 5.4	As a Finance Manager, I want to export financial summary reports so that accounting records remain up to date.

6. Customer Experience Officer (Ms. Tharushi Fernando)

User Story ID	User Story
US 6.1	As a Customer Experience Officer, I want to receive passenger complaints so that service issues can be investigated promptly.
US 6.2	As a Customer Experience Officer, I want to inspect ride histories and cancellation details so that cancellation fee disputes are resolved fairly.
US 6.3	As a Customer Experience Officer, I want to review passenger star ratings and driver reviews so that low-quality service can be flagged.
US 6.4	As a Customer Experience Officer, I want to send formal resolution responses to users so that their support tickets are officially resolved.

Product Backlog

The backlog is ordered first by business priority and then by dependency. High denotes core, security, transaction, or safety-critical capability; Medium denotes essential supporting capability; and Low denotes recovery or reporting enhancements that follow the operational core.

Item ID	Module	User Story Summary	Priority	Target Sprint
PBI-01	UserAccounts	US 1.1: As a Daily Passenger, I want to create an account and log in securely so that I can access ride-booking services.	High	Sprint 1
PBI-02	Ride Booking	US 1.2: As a Daily Passenger, I want to enter my pickup and drop-off points to view estimated fares before booking.	High	Sprint 1
PBI-03	Trip Request	US 1.3: As a Driver, I want to receive real-time ride notifications so that I can accept or decline incoming trip requests.	High	Sprint 1
PBI-04	User Accounts	US 2.1: As a Senior Driver Coordinator, I want to inspect driver registration applications so that I can ensure only qualified drivers join the fleet.	High	Sprint 1
PBI-05	User Accounts	US 2.2: As a Senior Driver Coordinator, I want to verify driver license and vehicle documents so that platform compliance is strictly maintained.	High	Sprint 1
PBI-06	Administration	US 3.1: As a System Admin, I want to configure role-based permissions so that users access features relevant to their roles.	High	Sprint 1
PBI-07	Trip Request	US 1.4: As a Driver, I want to update trip progress (Arrived, Started, Completed) so that the system logs my ride activity.	High	Sprint 2
PBI-08	Ride Booking	US 1.5: As a Daily Passenger, I want to track my assigned driver in real-time.	High	Sprint 2
PBI-09	Payments	US 1.6: As a Daily Passenger, I want to complete multi-channel payments and receive digital receipts.	High	Sprint 2
PBI-10	Administration	US 3.2: As a System Admin, I want to monitor real-time system activity dashboards so that I can ensure high application availability.	High	Sprint 2
PBI-11	Administration	US 3.4: As a System Admin, I want to view system audit logs so that platform activity can be analyzed during technical reviews.	Medium	Sprint 2
PBI-12	Payments	US 5.1: As a Finance Manager, I want to review transaction logs across card, digital wallet, and cash payments so that financial records are complete.	Medium	Sprint 2
PBI-13	Feedback	US 6.1: As a Customer Experience Officer, I want to receive passenger complaints so that service issues can be investigated promptly.	Medium	Sprint 3
PBI-14	Feedback	US 6.2: As a Customer Experience Officer, I want to inspect ride histories and cancellation details so that cancellation fee disputes are resolved fairly.	Medium	Sprint 3

PBI-15	Feedback	US 6.3: As a Customer Experience Officer, I want to review passenger star ratings and driver reviews so that low-quality service can be flagged.	Medium	Sprint 3
PBI-16	Feedback	US 6.4: As a Customer Experience Officer, I want to send formal resolution responses to users so that their support tickets are officially resolved.	Medium	Sprint 3
PBI-17	Administration	US 3.3: As a System Admin, I want to suspend accounts that commit severe policy violations so that security standards remain high.	Medium	Sprint 3
PBI-18	Payments	US 5.3: As a Finance Manager, I want to process approved refund requests for failed or disputed transactions so that client accounts stay balanced.	Medium	Sprint 3
PBI-19	Administration	US 4.1: As an Operations Manager, I want to view daily trip demand statistics so that regional ride allocation can be planned effectively.	Low	Sprint 4
PBI-20	Operations	US 4.2: As an Operations Manager, I want to track active driver availability trends during peak commuting hours so that service gaps are minimized.	Low	Sprint 4
PBI-21	Operations	US 4.3: As an Operations Manager, I want to inspect overall trip cancellation rates so that operational bottlenecks can be resolved.	Low	Sprint 4
PBI-22	Operations	US 4.4: As an Operations Manager, I want to generate operational reports so that performance insights can be shared with management.	Low	Sprint 4
PBI-23	User Accounts	US 2.3: As a Senior Driver Coordinator, I want to review driver performance metrics and ratings so that high-performing drivers are recognized.	Low	Sprint 4
PBI-24	User Accounts	US 2.4: As a Senior Driver Coordinator, I want to monitor repeated trip cancellations by drivers so that appropriate operational warnings can be issued.	Low	Sprint 4

Priority Rationale and Distribution

Priority	Count	Selection Rule	Planned Position
High	10	Core journey, access, safety, or transaction control	Sprint 1 PBI 1 - Sprint 2 PBI 10
Medium	8	Essential dependent operational capability	Sprint 2 PBI 11 - Sprint 3 PBI 18
Low	6	Deferrable recovery or reporting enhancement	Sprint 3 PBI 19 – Sprint 4 PBI 24

Sprint Backlog

The Product Backlog has been divided into 4 Sprints , Prioritizing High-value core functionalities first. Planned Sprint Task Assignments for all 6 Members implemented below.

Sprint 1 (Weeks 4–6):

Core Authentication & Ride Setup

Sprint Goal: Establish core user account access, driver registration verification, and initial passenger ride estimation capabilities to validate the foundational platform workflow.

Dependency order: User registration and authentication enable driver profile verification; verified driver setup enables initial ride booking requests and fare calculation.

Sprint ID	PBI	User Story	Priority
SP1-01	PBI-01	As a Daily Passenger, I want to create an account and log in securely so that I can access ride-booking services	High
SP1-02	PBI-02	As a Daily Passenger, I want to enter my pickup and drop-off points to view estimated fares before booking.	High
SP1-03	PBI-03	As a Driver, I want to receive real-time ride notifications so that I can accept or decline incoming trip requests.	High
SP1-04	PBI-04	As a Senior Driver Coordinator, I want to inspect driver registration applications so that I can ensure only qualified drivers join the fleet.	High
SP1-05	PBI-05	As a Senior Driver Coordinator, I want to verify driver license and vehicle documents so that platform compliance is strictly maintained.	High
SP1-06	PBI-06	As a System Admin, I want to configure role-based permissions so that users access features relevant to their roles.	High

Task Assignments and Ownership

Task ID	User Story	Task Description	Assigned Developer	Est.Hours
T1.1	PBI-01	Design registration & login UI screens	Nayanamina A.R.L.P.	6 hrs
T1.2	PBI-01	Implement JWT authentication & password recovery	Nayanamina A.R.L.P.	8 hrs
T1.3	PBI-02	Build pickup/destination map selector UI	Dharmakeerthi W.A.C.B.	6 hrs
T1.4	PBI-02	Implement fare estimation calculation endpoint	Dharmakeerthi W.A.C.B.	6 hrs
T1.5	PBI-03	Build real-time request pop-up interface for drivers	Senaka K.A.T.	7 hrs
T1.6	PBI-03	Implement Accept/Decline request handling logic	Senaka K.A.T.	6 hrs
T1.7	PBI-04	Build document upload and review API for driver verification	Nayanamina A.R.L.P.	5 hrs
T1.8	PBI-05	Implement driver document verification management panel	Nayanamina A.R.L.P.	5 hrs
T1.9	PBI-06	Implement RBAC security middleware for system routes	Rammandalagedara R.V.S.	6 hrs

Sprint 2 (Weeks 7–8):

Active Trip Execution & Financial Processing

Sprint Goal: Complete active ride dispatch workflows, live trip execution, multi-channel payment processing, and system audit monitoring.

Dependency order: Drivers accepting ride dispatch requests enables live trip updates and vehicle tracking; trip completion enables multi-channel payment processing, transaction logging, and administrative audit monitoring.

Sprint ID	PBI	User Story	Priority
SP2-01	PBI-07	As a Driver, I want to update trip progress (Arrived, Started, Completed) so that the system logs my ride activity.	High
SP2-02	PBI-08	As a Daily Passenger, I want to track my assigned driver's status and vehicle details so that I know when my ride will arrive.	High
SP2-03	PBI-09	As a Daily Passenger, I want to pay via card, wallet, or cash and receive a digital receipt so that I have proof of payment.	High
SP2-04	PBI-10	As a System Admin, I want to monitor real-time system activity dashboards so that I can ensure high application availability.	High
SP2-05	PBI-11	As a System Admin, I want to view system audit logs so that platform activity can be analyzed during technical reviews.	Medium
SP2-06	PBI-12	As a Finance Manager, I want to review transaction logs across card, digital wallet, and cash payments so that financial records are complete.	Medium

Task Assignments and Ownership

Task ID	User Story	Task Description	Assigned Developer	Est.Hours
T2.1	PBI-07	Build status update controls (Arrived, Started, Completed)	Senaka K.A.T.	5 hrs
T2.2	PBI-08	Implement vehicle location tracking UI component	Dharmakeerthi W.A.C.B.	8 hrs
T2.3	PBI-09	Integrate payment gateway for online card/wallet processing	Edirisinghe E.A.R.N.D.	10 hrs
T2.4	PBI-09	Build automated digital receipt PDF generation module	Edirisinghe E.A.R.N.D.	6 hrs
T2.5	PBI-10	Build real-time admin monitoring dashboard UI	Rammandalagedara R.V.S.	7 hrs
T2.6	PBI-11	Implement backend audit logging logger and viewer	Rammandalagedara R.V.S.	5 hrs
T2.7	PBI-12	Build payment transaction history and log view for finance	Edirisinghe E.A.R.N.D.	6 hrs

Sprint 3 (Weeks 9–10):

Quality Control, Complaints & Governance

Sprint Goal: Implement post-trip customer feedback collection, cancellation dispute resolution, financial refunds, and account policy enforcement.

Dependency order: Completed trips generate rating and complaint logs; logged complaints support customer support investigation, which in turn triggers formal ticket resolutions, refund processing, and administrative account suspensions.

Sprint ID	PBI	User Story	Priority
SP3-01	PBI-13	As a Customer Experience Officer, I want to receive passenger complaints so that service issues can be investigated promptly.	Medium
SP3-02	PBI-14	As a Customer Experience Officer, I want to inspect ride histories and cancellation details so that cancellation fee disputes are resolved fairly.	Medium
SP3-03	PBI-15	As a Customer Experience Officer, I want to review passenger star ratings and driver reviews so that low-quality service can be flagged.	Medium
SP3-04	PBI-16	As a Customer Experience Officer, I want to send formal resolution responses to users so that their support tickets are officially resolved.	Medium
SP3-05	PBI-17	As a System Admin, I want to suspend accounts that commit severe policy violations so that security standards remain high.	Medium
SP3-06	PBI-18	As a Finance Manager, I want to process approved refund requests for failed or disputed transactions so that client accounts stay balanced.	Medium

Task Assignments and Ownership

Task ID	User Story	Task Description	Assigned Developer	Est.Hours
T3.1	PBI-13	Create customer support ticketing and complaint logging interface	Weerasingha W.A.M.B.	6 hrs
T3.2	PBI-14	Build cancellation history and dispute inspection module	Weerasingha W.A.M.B.	5 hrs
T3.3	PBI-15	Build 1-5 star rating and written review submission modal	Weerasingha W.A.M.B.	5 hrs
T3.4	PBI-16	Implement support ticket resolution response notification module	Weerasingha W.A.M.B.	4 hrs
T3.5	PBI-17	Build account suspension and policy enforcement panel	Rammandalagedara R.V.S.	5 hrs
T3.6	PBI-18	Implement backend refund processing and commission calculations	Edirisinghe E.A.R.N.D.	7 hrs

Sprint 4 (Weeks 11–12):

Operations Dashboards & System Refinement

Sprint Goal: Deliver operational analytics, financial summary exports, performance reviews, and driver cancellation monitoring.

Dependency order: System operational logs enable demand analytics and peak-hour availability tracking; financial logs support report exports; historical trip ratings enable driver performance metric reviews and cancellation warnings.

Sprint ID	PBI	User Story	Priority
SP4-01	PBI-19	As an Operations Manager, I want to view daily trip demand statistics so that regional ride allocation can be planned effectively.	Low
SP4-02	PBI-20	As an Operations Manager, I want to track active driver availability trends during peak commuting hours so that service gaps are minimized.	Low
SP4-03	PBI-21	As an Operations Manager, I want to inspect overall trip cancellation rates so that operational bottlenecks can be resolved.	Low
SP4-04	PBI-22	As an Operations Manager, I want to generate operational reports so that performance insights can be shared with management.	Low
SP4-05	PBI-23	As a Senior Driver Coordinator, I want to review driver performance metrics and ratings so that high-performing drivers are recognized.	Low
SP4-06	PBI-24	As a Senior Driver Coordinator, I want to monitor repeated trip cancellations by drivers so that appropriate operational warnings can be issued.	Low

Task Assignments and Ownership

Task ID	User Story	Task Description	Assigned Developer	Est.Hours
T4.1	PBI-19	Build operations demand and trip volume analytics charts	Rammandalagedara R.V.S.	8 hrs
T4.2	PBI-20	Implement peak hour driver availability heatmap UI	Rammandalagedara R.V.S.	6 hrs
T4.3	PBI-21	Build cancellation rate analytics and breakdown view	Rammandalagedara R.V.S.	5 hrs
T4.4	PBI-22	Build financial and operational report export features (CSV/PDF)	Edirisinghe E.A.R.N.D.	6 hrs
T4.5	PBI-23	Build driver rating and performance review dashboard	Nayanamina A.R.L.P.	6 hrs
T4.6	PBI-24	Build driver repeated cancellation warning & flagging system	Nayanamina A.R.L.P.	5 hrs
T4.7	ALL	Perform full system integration testing and bug fixing	All Team Members	12 hrs

Member Ownership Summary

Member	Role	Owned Task IDs	Total Est. Hours
Nayanamina A.R.L.P.	Scrum Master	T1.1, T1.2, T1.7, T1.8, T4.5	30
Dharmakeerthi W.A.C.B.	Developer	T1.3, T1.4, T2.2	20
Senaka K.A.T.	Developer	T1.5, T1.6, T2.1	18
Edirisinghe E.A.R.N.D.	Developer	T2.3, T2.4, T2.7, T3.6, T4.4	35
Weerasingha W.A.M.B.	Developer	T3.1, T3.2, T3.3, T3.4	20
Rammandalagedara R.V.S.	Product Owner	T1.9, T2.5, T2.6, T3.5, T4.1, T4.2, T4.3	42
All Team Members	All	T4.7 (Shared System Integration & Testing)	12

Estimated hours are a planning aid. Sprint progress should be reviewed during Daily Scrum and adjusted transparently when new dependencies or risks appear.